

MATHEMATICAL ANALYSIS / ACHIEVEMENT SETS / SUMSETS

THE CONSTANT 3 IS SHARP

A middle-third witness under the literal formulation of Problem 13

SHARPNESS WITNESS

$$x_n = \frac{2}{3^n}, \quad r_n = \frac{1}{3^n}, \quad \mathcal{E}(x) + \mathcal{E}(x) = [0, 2]$$

MIDDLE-THIRD CANTOR SET C

SECOND COPY OF C

C + C = [0,2]

NO GAPS REMAIN

A Note on the Literal Formulation of Problem 13 for Fast Achievement-Set Sumsets

A middle-third witness under the literal formulation of Problem 13

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Research status

This manuscript gives a complete elementary argument for the literal formulation of Problem 13 in the cited survey. It has not undergone independent mathematical review or peer review. Because the counterexample is exceptionally simple, confirmation of the intended wording and a targeted prior-art check are particularly important before the result is represented as an established resolution.

Abstract

A 2026 survey article records the following question. If $\sum x_n$ is a fast convergent positive series with remainder $r_n = \sum_{k>n} x_k$, a stated sufficient condition for $\mathcal{E}(x) + \mathcal{E}(x)$ to be a Cantor set of Lebesgue measure zero is $3^n r_n \rightarrow 0$. Can the constant 3 be replaced by a smaller number? We answer the literal question negatively. For the normalized middle-third sequence

$$x_n = \frac{2}{3^n},$$

one has $\sum x_n = 1$, $r_n = 3^{-n}$, and $x_n = 2r_n$, so the series is fast convergent. For every fixed $0 < c < 3$,

$$c^n r_n = \left(\frac{c}{3}\right)^n \rightarrow 0.$$

Nevertheless, $\mathcal{E}(x)$ is the standard ternary Cantor set C and $C + C = [0, 2]$. Thus every replacement $c < 3$ admits an interval-valued counterexample, and 3 is optimal. We also show that the same digit argument proves a general sharpness principle: for each integer $m \geq 2$, the sequence $x_n = m/(m+1)^n$ is fast convergent and its m -fold achievement-set sum is $[0, m]$, showing that the natural exponential constant $m+1$ cannot be lowered.

Main conclusion. For every real constant c with $0 < c < 3$, there exists a fast convergent positive series of sum 1 such that $c^n r_n \rightarrow 0$, while the self-sum of its achievement set is the full interval $[0, 2]$. Hence the constant 3 in the stated criterion is sharp and cannot be replaced by any smaller positive constant.

MSC 2020. Primary 40A05; Secondary 11B05, 28A80.

Keywords. Achievement set, Cantor set,

Minkowski sum, fast convergent series, sharp constant.

1 Introduction

For a summable sequence $x = (x_n)$ of positive real numbers, its *achievement set* is

$$\mathcal{E}(x) = \left\{ \sum_{n=1}^{\infty} \varepsilon_n x_n : \varepsilon_n \in \{0, 1\} \right\}.$$

Write

$$r_n = \sum_{k>n} x_k$$

for the remainder after the n th term. The series is called *fast convergent* when $x_n > r_n$ for every n . In that case $\mathcal{E}(x)$ is a central Cantor set; see, for example, Jones [2] and Nitecki [4].

Głąb and Prus-Wiśniowski report the following sufficient condition in Theorem 12 of their survey article [1]: if $\sum x_n$ is fast convergent, has sum 1, and

$$\lim_{n \rightarrow \infty} 3^n r_n = 0,$$

then $\mathcal{E}(x) + \mathcal{E}(x)$ is a Cantor set of Lebesgue measure zero. Immediately afterward they pose:

Problem 13 in the cited survey

Can the constant 3 in the first implication be replaced by a smaller number?

The middle-third Cantor set itself gives a negative answer. The underlying reason is exact: the base 3 in the remainder test is the reciprocal of the geometric ratio at which the three sum digits first close every gap.

Interpretive caveat

The argument below addresses the printed formulation of Problem 13. Since the witness is classical, it is possible that the survey intended an additional hypothesis, a different quantifier, or a different direction of optimization. External review should first confirm the intended statement with the survey authors.

2 Preliminaries

For compact sets $A, B \subset \mathbb{R}$, their Minkowski sum is

$$A + B = \{a + b : a \in A, b \in B\}.$$

For a positive integer m , write $mA = A + \dots + A$ for the m -fold sum.

We use only the elementary positional-expansion lemma below.

Lemma 2.1 (Base-expansion lemma). *Let $b \geq 2$ be an integer. Every $u \in [0, 1]$ admits an expansion*

$$u = \sum_{n=1}^{\infty} \frac{d_n}{b^n}, \quad d_n \in \{0, 1, \dots, b-1\}.$$

Proof. For $u = 1$, take $d_n = b - 1$ for all n . Suppose $0 \leq u < 1$. Set $t_0 = u$ and recursively define

$$d_n = \lfloor b t_{n-1} \rfloor, \quad t_n = b t_{n-1} - d_n.$$

Then $d_n \in \{0, \dots, b-1\}$ and $t_n \in [0, 1)$. Iteration gives

$$u = \sum_{j=1}^n \frac{d_j}{b^j} + \frac{t_n}{b^n}.$$

The remainder tends to zero, proving the claim. □

3 The middle-third witness

Define

$$x_n = \frac{2}{3^n}, \quad n \geq 1. \quad (1)$$

Proposition 3.1. *The sequence in equation (1) has total sum 1, remainder*

$$r_n = \frac{1}{3^n},$$

and satisfies $x_n = 2r_n$. In particular, it is fast convergent.

Proof. The geometric-series identities give

$$\sum_{n=1}^{\infty} \frac{2}{3^n} = 1$$

and

$$r_n = \sum_{k=n+1}^{\infty} \frac{2}{3^k} = \frac{2/3^{n+1}}{1 - 1/3} = \frac{1}{3^n}.$$

Thus $x_n = 2r_n > r_n$. □

The corresponding achievement set is

$$\mathcal{E}(x) = \left\{ \sum_{n=1}^{\infty} \frac{2\varepsilon_n}{3^n} : \varepsilon_n \in \{0, 1\} \right\},$$

which is precisely the standard middle-third Cantor set $C \subset [0, 1]$.

Theorem 3.2. *For the sequence in equation (1),*

$$\mathcal{E}(x) + \mathcal{E}(x) = C + C = [0, 2].$$

Proof. The inclusion $C + C \subset [0, 2]$ is immediate. Conversely, let $t \in [0, 2]$ and set $u = t/2 \in [0, 1]$. By lemma 2.1, choose ternary digits $d_n \in \{0, 1, 2\}$ such that

$$u = \sum_{n=1}^{\infty} \frac{d_n}{3^n}.$$

For each n , choose $\varepsilon_n, \eta_n \in \{0, 1\}$ with

$$d_n = \varepsilon_n + \eta_n;$$

this is possible for every digit 0, 1, 2. Then

$$t = 2u = \sum_{n=1}^{\infty} \frac{2\varepsilon_n}{3^n} + \sum_{n=1}^{\infty} \frac{2\eta_n}{3^n},$$

and the two series on the right represent points of C . Hence $t \in C + C$. □

stage-5 approximation to C



a second copy of C



the three first-level sum channels have adjacent convex hulls

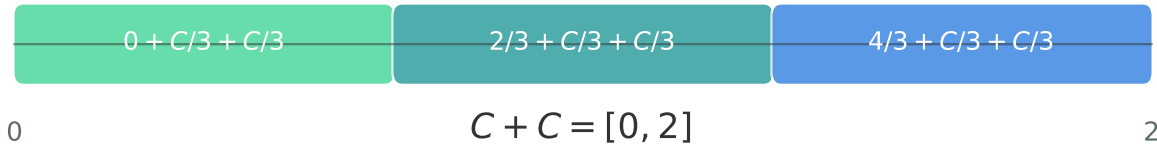


Figure 1: Two stage-5 Cantor approximations and the three first-level sum channels. Their convex hulls meet at $2/3$ and $4/3$; the digit proof in [theorem 3.2](#) shows that the full recursive sum contains every point of $[0, 2]$.

4 Sharpness of the constant

Theorem 4.1 (Sharpness theorem). *Let c be any real number with $0 < c < 3$. There exists a fast convergent positive series $\sum x_n$ of sum 1 such that*

$$c^n r_n \rightarrow 0,$$

but $\mathcal{E}(x) + \mathcal{E}(x)$ is an interval. Consequently, under the literal stated sufficient condition, the constant 3 cannot be replaced by any smaller positive constant.

Proof. Use the single sequence $x_n = 2/3^n$, independently of c . By [proposition 3.1](#), $r_n = 3^{-n}$, and therefore

$$c^n r_n = \left(\frac{c}{3}\right)^n \rightarrow 0$$

because $c < 3$. By [theorem 3.2](#), however,

$$\mathcal{E}(x) + \mathcal{E}(x) = [0, 2],$$

which is not a Cantor set and does not have measure zero. □

Corollary 4.2. *Under the literal printed wording quoted in this manuscript, the proposed reduction of the constant is refuted by [theorem 4.1](#). This is a statement about that wording, not a claim that the survey authors intended no additional restriction.*

The example lies exactly on the boundary of the original hypothesis:

$$3^n r_n = 1 \quad \text{for every } n.$$

Thus it does not contradict the stated theorem. It shows instead that any reduction of the exponential base immediately admits the interval-producing boundary case.

4.1 The geometric phase transition

For additional context, fix $q \in (0, 1/2)$ and consider the normalized geometric series

$$x_n = (1 - q)q^{n-1}.$$

Its sum is 1, its remainder is $r_n = q^n$, and it is fast convergent because

$$x_n - r_n = q^{n-1}(1 - 2q) > 0.$$

Its self-sum is the compact set

$$K_q = \left\{ \sum_{n=1}^{\infty} d_n(1 - q)q^{n-1} : d_n \in \{0, 1, 2\} \right\}.$$

Proposition 4.3. For $0 < q < 1/2$:

- (i) if $q < 1/3$, then K_q is a Cantor set;
- (ii) if $q \geq 1/3$, then $K_q = [0, 2]$.

Proof. At the first digit, the three possible channels have convex hulls

$$I_0 = [0, 2q], \quad I_1 = [1 - q, 1 + q], \quad I_2 = [2 - 2q, 2].$$

If $q < 1/3$, adjacent intervals are disjoint. Repeating the same separation at every scale yields a compact perfect totally disconnected set, hence a Cantor set.

If $q \geq 1/3$, the intervals cover $[0, 2]$. Given $t_0 \in [0, 2]$, choose $d_1 \in \{0, 1, 2\}$ such that $t_0 \in I_{d_1}$ and set

$$t_1 = \frac{t_0 - d_1(1 - q)}{q} \in [0, 2].$$

Repeat this construction. After n steps,

$$t_0 = \sum_{j=1}^n d_j(1 - q)q^{j-1} + q^n t_n.$$

Since $0 \leq t_n \leq 2$ and $q^n \rightarrow 0$, the limiting expansion places t_0 in K_q . Thus $K_q = [0, 2]$. □

For this family,

$$3^n r_n = (3q)^n.$$

The condition $3^n r_n \rightarrow 0$ is therefore exactly $q < 1/3$, the Cantor side of the transition in [proposition 4.3](#). Replacing 3 by any $c < 3$ moves the analytic threshold to $q < 1/c$, thereby including $q = 1/3$, where the sumset is already an interval.

5 A general m -fold sharpness principle

The same construction is not special to two-fold sums.

Theorem 5.1. Let $m \geq 2$ be an integer and define

$$x_n = \frac{m}{(m + 1)^n}.$$

Then $\sum x_n = 1$, the series is fast convergent with

$$r_n = \frac{1}{(m + 1)^n},$$

and

$$m\mathcal{E}(x) = [0, m].$$

Consequently, for every $0 < c < m + 1$ one has $c^n r_n \rightarrow 0$ although the m -fold sum is an interval.

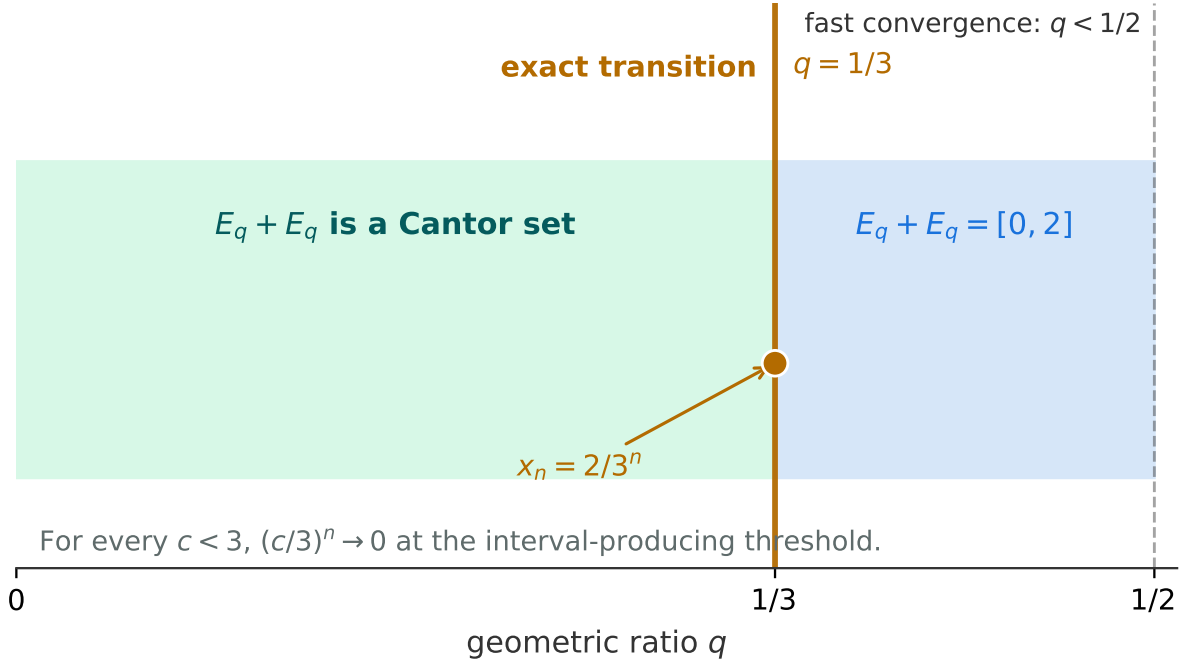


Figure 2: The normalized geometric family has a precise self-sum transition at $q = 1/3$. The witness $x_n = 2/3^n$ sits at that boundary, while every proposed base $c < 3$ satisfies $(c/3)^n \rightarrow 0$.

Proof. The geometric-series calculation gives

$$\sum_{n=1}^{\infty} \frac{m}{(m+1)^n} = 1, \quad r_n = \frac{1}{(m+1)^n},$$

so $x_n = mr_n > r_n$.

Every point of $m\mathcal{E}(x)$ has the form

$$\sum_{n=1}^{\infty} \frac{md_n}{(m+1)^n}, \quad d_n \in \{0, 1, \dots, m\},$$

because d_n counts how many of the m selected subsums use the n th term. Hence $m\mathcal{E}(x) \subset [0, m]$. Conversely, let $t \in [0, m]$. Apply [lemma 2.1](#) to $t/m \in [0, 1]$ in base $m+1$:

$$\frac{t}{m} = \sum_{n=1}^{\infty} \frac{d_n}{(m+1)^n}, \quad d_n \in \{0, \dots, m\}.$$

For each n , express d_n as the sum of m binary digits. Grouping those digits by copy writes t as a sum of m points of $\mathcal{E}(x)$. Therefore $m\mathcal{E}(x) = [0, m]$. Finally,

$$c^n r_n = \left(\frac{c}{m+1} \right)^n \rightarrow 0$$

whenever $c < m+1$. □

Remark 5.2. The number $m+1$ is the cardinality of the digit set $\{0, 1, \dots, m\}$ appearing in an m -fold sum. The exact witness in [theorem 5.1](#) shows that the familiar cylinder-cover base $m+1$ is not merely an artifact of a rough estimate; it is attained at the interval threshold.

At $q = 1/(m + 1)$, the $m + 1$ digit channels tile $[0, m]$ exactly (shown for $m = 5$).

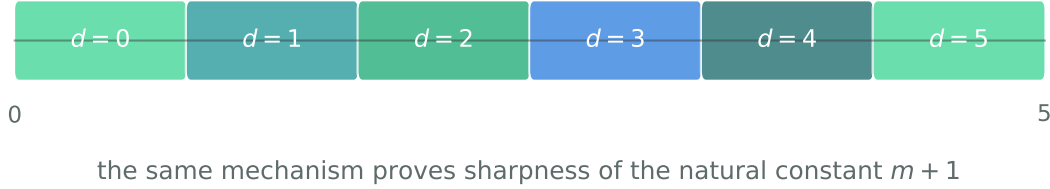


Figure 3: The generalized boundary construction. At $q = 1/(m + 1)$, the $m + 1$ digit channels have consecutive convex hulls and the base- $(m + 1)$ digit expansion fills $[0, m]$.

6 Discussion and verification priorities

The proof of [theorem 4.1](#) uses only three identities:

$$\sum_{n \geq 1} \frac{2}{3^n} = 1, \quad r_n = \frac{1}{3^n}, \quad C + C = [0, 2].$$

The first two are geometric-series calculations. The third follows from ternary expansion and digitwise decomposition $d_n = \varepsilon_n + \eta_n$.

The mathematical argument is therefore short, but the research-status question is not. Before this note is characterized as answering the intended open problem, reviewers and the survey authors should determine:

1. whether Problem 13 is intended exactly as printed in the survey;
2. whether the survey uses the standard remainder convention $r_n = \sum_{k > n} x_k$ in this statement;
3. whether an unstated restriction was intended to exclude the boundary geometric example;
4. whether the observation already appears in correspondence, unpublished notes, or later work not indexed by the literature search conducted for this preprint.

Independent review target

A reviewer can check the entire mathematical core by verifying [proposition 3.1](#) and [theorems 3.2](#) and [4.1](#). The remaining issue is primarily interpretation and novelty, not a long chain of technical estimates.

Reproducibility statement

The source package accompanying this preprint contains the complete Lua^AT_EX manuscript, formula-derived figures, and an exact-arithmetic verification script. The script checks the remainder formulas, fast-convergence inequalities, finite base-expansion coverage, and the generalized m -fold tiling identities. No numerical approximation is used in the proof.

Institutional disclaimer and computational research declaration

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Generative artificial intelligence and other computational tools were used as research instruments to support construction search, symbolic and numerical analysis, literature review, proof development, verification, and manuscript preparation. Their use does not constitute independent validation of the results presented.

This manuscript has not yet undergone independent peer review. Its unconditional mathematical propositions should not be confused with confirmation that they answer the survey authors’ intended problem; that interpretation requires clarification from the source authors and independent examination by qualified mathematicians. Independent verification, criticism, replication, and identification of errors are expressly invited.

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Data and code availability

No external empirical dataset was used. The complete Lua $\text{\LaTeX}$ source, figures, verification code, recorded output, review guide, and release checksums are included in the source and reproducibility package associated with this candidate edition. Computational checks verify only the claims explicitly encoded by those programs and do not independently validate unencoded infinite arguments, interpretation, or novelty.

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Competing interests

Daniel H. Jeffery is the founder and Executive Director of Metriq Foundation, Inc. No other competing interests are declared.

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